

Jacob Cascone

AI ENGINEER

Mission, KS (Remote) | 816-520-5652 | cobo.cascone@gmail.com | lessondraft.com | github.com/cascone26

SUMMARY

AI engineer who ships production products end-to-end by directing AI-assisted development at speed. Built and operate a live SaaS (LessonDraft) with real paying subscribers — full stack from Stripe billing to a multi-provider LLM generation pipeline with automatic failover. Strong full-stack range (Next.js/React/TypeScript, Postgres/Supabase, payment infrastructure) paired with hands-on LLM integration (Claude, prompt design, multi-provider fallback architecture, agent orchestration). Comfortable fully remote and self-directed — everything below was built and operated solo.

SKILLS

AI / LLM	Claude API, multi-provider LLM fallback design, prompt engineering, agent orchestration, RAG-adjacent retrieval/grounding	Frontend	Next.js, React, TypeScript, Tailwind CSS
Backend / Data	Node.js, Postgres, Supabase, REST APIs, auth (NextAuth/Clerk)	Infra / Payments	Vercel deployment, Stripe billing/webhooks, production monitoring & health checks, cross-machine failover
Dev Workflow	AI-assisted development (Claude Code), Git/GitHub, root-cause debugging	Work Style	Fully remote (full-time, part-time, or contract), self-directed, ships solo end-to-end

FEATURED PROJECTS

LessonDraft — AI Lesson Planning Platform for K-12 Teachers 2025 – Present
lessondraft.com · Solo builder & operator · Live, real paying subscribers

- Built a full production SaaS solo: Next.js frontend, Stripe subscription billing (3 tiers), Supabase-backed accounts, and 40+ AI-generated tools (lesson plans, worksheets, rubrics, IEP goals, report-card comments)
- Designed a multi-provider AI generation pipeline with automatic failover across free-tier, self-hosted, and premium models — kept marginal AI cost near zero with self-healing health checks that route around a dead provider automatically
- Audited full Stripe billing surface end-to-end against live checkout data; found and fixed 4 independent revenue-blocking bugs in one pass, including a mispriced plan that had been overcharging silently for 5 months and a webhook gap that would have let failed-payment customers keep paid access indefinitely
- Built custom production monitoring (synthetic "customer journey" canaries, health checks) and a cross-machine backup/disaster-recovery path for the AI generation layer

Meta Tutor — Multi-Subject AI Learning Platform 2026 – Present
Solo builder & operator

- Refactored a single-subject study tool into a subject-agnostic architecture — one shared AI-chat layer, progress-tracking interface, and content-grounding registry, with each new subject (8 real classes, chess, trivia) plugging in as an adapter instead of duplicated code
- Built a chess module integrating a real chess engine client-side (zero server cost), with move-quality analysis wired into the same cross-subject progress system
- Diagnosed and fixed 2 silent production incidents via live investigation (a corrupted environment variable that silently broke every database feature app-wide; a stale auth-token sync gap breaking all AI routes)

Multi-Round AI Verification Pipeline for a 36K-Asset Catalog 2026
Personal game-asset project

- Designed a multi-round verification pipeline (31-agent contact-sheet review, then an adversarial "skeptic" AI pass, then personal human spot-check at full resolution) to QA a 36,000+ sprite catalog against intended name/appearance — found that even the adversarial AI layer had real error rates in both directions, closed only by human-in-the-loop review; final verified set hard-blocked in the generation pipeline

OPEN SOURCE

✓ **llm-ladder** — multi-provider LLM failover library, extracted from LessonDraft's production system; chaos-tested against injected timeouts/rate-limits/malformed responses, 10/10 passing

✓ **deep-down** — 6-phase Roblox game, engine-verified via automated headless testing

✓ **standards-rag** — RAG over real Common Core standards; from-scratch BM25 retrieval + honest precision eval (70%@5)

✓ **agentic-grader** — agentic grading pipeline w/ human-approval gate; low-confidence grades never auto-commit, 8/8 tests

EXPERIENCE

6th Grade Lead Teacher 2024 – Present
Regina Caeli Academy — Lenexa, KS

- Teach core subjects to 6th graders while building AI tools (LessonDraft, Meta Tutor) directly informed by real classroom needs — the same problems I build products to solve
- Assess and grade student work daily using structured rubrics; deliver precise, actionable written feedback

After-School Care Assistant 2024 – Present
Visitation Catholic School — Kansas City, MO

EDUCATION

Grand Canyon University — Phoenix, AZ (Online) B.A. in Education · In Progress
St. James Academy — Lenexa, KS High School Diploma · 05/2023